

Exotic Triangulations and the Octahedral Axiom

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Homotopy category and derived category

Many constructions over morphisms of cochain complexes only depend on the homotopy class, so it makes sense to work in the (smaller) homotopy category

$$\mathbf{K}(\mathcal{A}) := \mathbf{C}(\mathcal{A}) / \simeq .$$

Another standard construction is the derived category $\mathbf{D}(\mathcal{A})$, which inverts the homology isomorphisms.

However, neither of those categories is abelian. But they do have a triangulated structure!

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What is a triangulated category? I

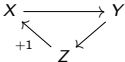
Definition

A *triangulated category* is an additive category $\mathcal{T}$ together with an autoequivalence $\Sigma: \mathcal{T} \rightarrow \mathcal{T}$ and a class of distinguished triangles

$$X \rightarrow Y \rightarrow Z \rightarrow \Sigma X$$

satisfying some axioms (TR1)-(TR4).

Abelian vs triangulated categories

Abelian category	Triangulated category
Is a property of the category	Is some extra structure
Short exact sequences $X \hookrightarrow Y \twoheadrightarrow Z$	Distinguished triangles 
Short exact sequences induce long exact sequences in 'derived functors'	Distinguished triangles induce long exact sequences after applying representable or 'cohomological' functors
First isomorphism theorem: $Z \cong Y/X$	Z only depends on $X \rightarrow Y$ up to (non-canonical) isomorphism

Classification of triangulated categories

Derived categories are not the only widely used examples of triangulated categories. We could make the following division:

- **Algebraic:** those that are equivalent to the stable category of a Frobenius category $\underline{\mathcal{F}}$ (e.g. $\mathbf{K}(\mathcal{A})$ and $\mathbf{D}(\mathcal{A})$).
- **Topological:** those that are equivalent to the homotopy category of a stable model category $ho\mathcal{M}$.
- **Exotic:** those that are neither algebraic nor topological, i.e. those that admit no enhancement.

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There are only a few known examples of exotic triangulated categories:

- Muro, Schwede and Strickland [MSS07] gave an exotic structure to the category of finitely generated projective modules over $\mathbb{Z}/4\mathbb{Z}$.
- Rizzardo and Van den Bergh [RV20] gave a class of exotic examples which are linear over fields of characteristic zero.

We have found a third class of examples which are linear over algebraically closed fields of any characteristic.

Our example: Underlying category

We consider the preprojective algebra of type L_2 over an algebraically closed field k . This is the path algebra

$$P(L_2) = k \left(b \curvearrowright 1 \begin{array}{c} \xrightarrow{a_1} \\ \xleftarrow{a_2} \end{array} 2 \right) / (b^2 + a_2 a_1, a_1 a_2).$$

If you are not familiar with path algebras, just think of it as the free algebra generated by elements

$$\{e_1, e_2, a_1, a_2, b\}$$

with the relations

$$\begin{aligned} e_1^2 &= e_1, & e_2^2 &= e_2, & e_1 e_2 &= 0 = e_2 e_1, \\ e_2 a_1 &= a_1 = a_1 e_1, & e_1 a_2 &= a_2 = a_2 e_2, & e_1 b &= b = b e_1, \\ b^2 + a_2 a_1 &= 0, & a_1 a_2 &= 0. \end{aligned}$$

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Our example: Underlying category

We consider the category of finitely generated modules over $P(L_2)$. This category is equivalent to the category of representations of $P(L_2)$, which has:

- Objects: Diagrams of f.d. vector spaces of the form

$$B \hookrightarrow V_1 \begin{matrix} \xrightarrow{A_1} \\ \xleftarrow{A_2} \end{matrix} V_2$$

which satisfy the relations $B^2 + A_2A_1 = 0$ and $A_1A_2 = 0$.

- Morphisms: Commutative diagrams of the form

$$\begin{array}{ccccc} B \hookrightarrow V_1 & \begin{matrix} \xrightarrow{A_1} \\ \xleftarrow{A_2} \end{matrix} & V_2 \\ \downarrow f_1 & & \downarrow f_2 \\ C \hookrightarrow W_1 & \begin{matrix} \xrightarrow{D_1} \\ \xleftarrow{D_2} \end{matrix} & W_2 \end{array}$$

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Our example: Underlying category

The underlying category of our example is the category of finitely generated projective modules over $P(L_2)$, whose objects are direct sums of the following two indecomposable representations:

$$P^1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \hookrightarrow k^4 \begin{array}{c} \xrightarrow{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}} k^2 \\ \xleftarrow{\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}} \end{array}$$

$$P^2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \hookrightarrow k^2 \begin{array}{c} \xrightarrow{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}} k^2 \\ \xleftarrow{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}} \end{array}$$

Our example: Shift functor

Our shift functor is given by twisting by the automorphism

$$\sigma: P(L_2) \rightarrow P(L_2), \quad a_1 \mapsto -a_1, \quad a_2 \mapsto -a_2, \quad b \mapsto -b - b^3.$$

Explicitly, it is the restriction of the functor

$$(-)_\sigma: \text{mod}(P(L_2)) \rightarrow \text{mod}(P(L_2))$$

defined by:

$$B \curvearrowright V_1 \begin{matrix} \xrightarrow{A_1} \\ \xleftarrow{A_2} \end{matrix} V_2 \mapsto -B-B^3 \curvearrowright V_1 \begin{matrix} \xrightarrow{-A_1} \\ \xleftarrow{-A_2} \end{matrix} V_2$$

$$(f_1, f_2) \mapsto (f_1, f_2).$$

Our example: Triangles

The distinguished triangles are sums of shifts of the following ones:

$$0 \rightarrow P^1 \xrightarrow{id} P^1 \rightarrow 0,$$

$$0 \rightarrow P^2 \xrightarrow{id} P^2 \rightarrow 0,$$

$$P_\sigma^1 \xrightarrow{b^3} P^1 \xrightarrow{\begin{pmatrix} b \\ -a_2 \end{pmatrix}} P^1 \oplus P^2 \xrightarrow{\begin{pmatrix} b & a_1 \end{pmatrix}} P^1,$$

$$P_\sigma^1 \xrightarrow{a_2 b} P^2 \xrightarrow{b a_1} P^1 \xrightarrow{b} P^1,$$

$$P_\sigma^1 \oplus P_\sigma^2 \xrightarrow{\begin{pmatrix} -b^2 & -b a_1 \\ a_2 b & 0 \end{pmatrix}} P^1 \oplus P^2 \xrightarrow{\begin{pmatrix} -b^2 & 0 \\ b & a_1 \end{pmatrix}} P^1 \oplus P^1 \xrightarrow{\begin{pmatrix} b & b^2 \\ -a_2 & 0 \end{pmatrix}} P^1 \oplus P^2,$$

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$$P_\sigma^2 \xrightarrow{-a_2 b a_1} P^2 \xrightarrow{a_1} P^1 \xrightarrow{a_2} P^2.$$

Idea of the construction

The idea comes from the following result:

Theorem (F. Muro, 2022, [Mur22])

Let Λ be a basic connected non-separable twisted-3-periodic f.d. algebra over a perfect field k . Then, there is an automorphism $\tau \in \text{Aut}(\Lambda)$ such that twisting by τ is essentially the unique shift functor $\Sigma = (-)_{\tau}: \text{proj}(\Lambda) \rightarrow \text{proj}(\Lambda)$ that makes $(\text{proj}(\Lambda), \Sigma)$ admit an algebraic triangulated structure.

So, the idea is to modify the automorphism τ into another automorphism σ such that $(-)_{\tau}$ is not isomorphic to $(-)_{\sigma}$ but in a way that $\text{proj}(\Lambda)$ still admits a pretriangulated structure with shift $(-)_{\sigma}$.

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Idea of the construction

In particular, the modification comes from composing with another automorphism κ which is a representative for a nontrivial element of the kernel of

$$Out(\Lambda) \rightarrow \underline{Pic}(\Lambda),$$

where

$$Out(\Lambda) = Aut(\Lambda)/Inn(\Lambda),$$

$$\underline{Pic}(\Lambda) := \{ \text{Triangulated self-equivalences of } \underline{mod}(\Lambda) \} / \cong_{tr},$$

and the map is given by $[\varphi] \mapsto [(-)_{\varphi}]$.

Theorem

Let Λ and τ be as in the previous theorem. Then, any nontrivial element κ in the kernel of $\text{Out}(\Lambda) \rightarrow \underline{\text{Pic}}(\Lambda)$ gives rise to a non-algebraic pretriangulated structure on $(\text{proj}(\Lambda), (-)_{\kappa \circ \tau})$.

The proof uses a classical theorem by A. Heller [Hel68].

Furthermore, our particular example can also be shown to be non-topological.

Bonus: The octahedral axiom

The octahedral axiom (TR4) is the axiom of triangulated categories which is an analogue of the third isomorphism theorem in abelian categories:

$$A \subseteq B \subseteq C \quad \rightsquigarrow \quad \frac{C/A}{B/A} \cong C/B.$$

Open problem: There is no known example of a pretriangulated category that does not satisfy the octahedral axiom, nor has it been proven to be a consequence of the other axioms (TR1)-(TR3).

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What we have tried

- Recover it from the octahedral axiom on the algebraic structure.
- Adapt the proof of the algebraic case.
- Try to disprove it by finding a counterexample by computer. We have not been able to find any, but the search is slow.
- Use a (computable) sufficient condition given in [Mur20] in terms of the differential of a spectral sequence. It turns out that it is not satisfied. It is also the first example where this condition has been checked with a negative result.

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- Give an equivalent condition for the octahedral axiom in terms of a secondary cohomology operation. We have been able to get this, but it is not clear whether or not it helps.
- **Classifying initial data:** We can try to classify all the possible initial data and check the axiom for each possible case. This is equivalent to classifying all morphisms of $\underline{mod}(\Lambda)$. We don't know how to do this, but it may be a path to follow.

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Thank you for your attention!